

Name: Team Timbuktu

Date: 3/3/25

Element L: Presentation of designer's recommendations

In line with the needs of our stakeholders, we compiled a list of explicit recommendations for enhancing the performance, safety, and reliability of the Eland toy crane based on practical engineering concepts, feedback from the stakeholders, and environmentally friendly and affordable options. We began by examining the existing design of the crane, the issues that confront the users, and the environment. We then had discussions to ensure that the improvements were aligned with the needs of the stakeholders. The ultimate recommendations comprised a crank mechanism, reinforcing the structural joints with T and L brackets, and stabilizing the crane using a guy-wire mechanism. The recommendations also had accompanying detailed designs to ensure that they are easily interpreted and implemented. As seen below, we elaborate on the purpose of each solution, how to implement them, and what results they are likely to achieve to present a comprehensive image of the recommended improvements.

The initial proposal was to make it simpler to use the crane with the addition of a hand-crank. Some people commented that it was difficult for younger individuals or those who are weak to pull the eland toy back across the fence manually. This would discourage them from using it and potentially break the toy or fence if they pulled with much force. To remedy that, we proposed a crank mechanism that converts circular motion to straight motion. This would allow individuals to pull the toy back effortlessly and smoothly. The crank is mounted on the horizontal beam of the crane and is connected to a spool with a sturdy cord or chain. When turned, the spool rotates and slowly brings back the toy. This makes the action simpler and provides better control so that there is no sudden pull that might damage the components.

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We carefully designed the crank based on considerations for the material, how comfortable it would be, and how much it would work effectively. We made the handle of the crank out of a light but strong aluminum alloy, which would be easy to turn and rugged. We roughened the spool so the string would have minimal slipping and made it large enough so a small pull would be effective. We attached the crank to the arm of the crane with rust-proof stainless steel screws and added a grommet made of a rubber material to dampen sound and vibration. We tested the prototypes extensively before mounting them to ensure they functioned effectively with various weights, and modified the gear ratio to correspond with the weight of the toy and the height of the fence. We also did user testing to determine how simple it would be to hold and hand-crank the handle, which resulted in a design adjustment to the grip shape and the inclusion of a ball-bearing mechanism within the crank housing. These enhancements ensured the finished product was easy to use and consistent, achieving the goal of being easy to use and strong.

The second proposal discussed issues with the crane, particularly where the horizontal arm dips or swings when it is out over the fence. Individuals involved observed that operating the crane over a long period or with heavy loads exacerbated the problem, which presented a safety hazard and reduced how long the crane functioned. To make the arm sturdier and support it being attached to the vertical post, we proposed adding T and L brackets on key joints. T brackets with a shape resembling the capital letter "T" would be put where the horizontal and vertical portions come together to make it sturdier, while L brackets that have a right-angle shape would keep the horizontal arm positioned, preventing it from bending sideways. The brackets would be constructed from galvanized steel, a sturdy material because it is not likely to rust, so they would last long even when exposed outdoors.

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Installing the brackets was a step-by-step affair for accuracy and durability. First, laser scanning produced a 3D image of the crane's present form, identifying stress areas and those that had easy bending. Based on that knowledge, the brackets were specifically designed to fit snugly over the crane's joints, with pre-drilled holes to correspond with the existing patterns so that they could easily be mounted. During assembly, the brackets were secured temporarily so that the engineers could test the crane's motion and carrying capacity before they were welded in place. To ensure the crane's original form could be protected, low-heat methods were used for welding on it, and a zinc-based primer was applied subsequently to prevent the welds from rusting. Thick gaskets made of rubber were also inserted between the surfaces of the crane and brackets to dampen vibration and reduce wear on the metal. Once mounted, the sturdy joints underwent simulated loads of extreme types, such as full extension and instantaneous side loads, to demonstrate their performance. This meticulous approach not only resolved the stability issue but also ensured that the crane had a longer lifespan, addressing the stakeholders' requirement for a low-maintenance, robust solution.

Third on the list was to prevent the crane from falling forward. This issue occurs when the horizontal arm is extended to its full length or when the loads are unbalanced. Individuals who were involved indicated that this could cause safety issues and injury to the toy or other structures around it. Our proposed solution was to repair it using a guy-wire stabilizing mechanism, a prevalent technique for stabilizing tall structures in engineering terms. The design was to install a sturdy cable made of steel near the top of the crane post and anchor it at a safe distance from the ground. The cable would balance out the pressure on the horizontal arm and stabilize the crane.

Installation of the guy-wire system required careful planning to adhere to the environment and safety guidelines for the user. The soil on the site was tested to identify

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the optimal depth and material for ground anchors. Helical anchors were selected as they gripped packed and loose soils well. The cable was purchased from a manufacturer of marine-grade stainless steel, which is not affected by rust and can cope with moisture and temperature variations. A turnbuckle was incorporated into the cable arrangement so that pressure could be tightened gently to accommodate soil changes or the settling of the building with seasons. To prevent tripping, the cable was encapsulated with a reflective UV-resistant plastic cover that is highly visible, and it was secured with a recessed plate on the ground surface. Regulations for inspection at regular intervals were established based on the safety standards, which involved tension checks at least every month and checking the anchors' strength every year, along with maintenance records stored in a logbook. A second system, a backup cable with a lesser tension rating, was also installed as a safety net, to be implemented only when the primary cable fails. This stabilization strategy addressed the threat of collapse immediately and reflected a clear emphasis on long-term security, which is similar to the primary aims of the stakeholders.

In brief, solutions presented to the stakeholders resulted from collaboration, blending new thoughts with practical application. By incorporating a crank mechanism, bracing crucial joints with specially made brackets, and implementing a two-layer guy-wire configuration, the proposed changes addressed each issue with strict requirements for being robust, safe, and economical. Instructions for implementing the changes incorporated material details, steps for testing, and maintenance guidelines, presenting a comprehensive source for conducting the work. This served to minimize disruptions to operations and guarantee a high return on investment. These improvements made the Eland toy crane a sturdy and simple-use system that could sustain heavy usage while safeguarding equipment and people. The feedback from the stakeholders reflected that

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they valued the meticulous and beneficial solutions, demonstrating mutual concern for quality and innovative thinking in achieving their objectives.